THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Sequences and Progressions** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 53	$2 \times 25 + 3$.
2	**(c)** 17th term	$3n + 2 = 53$ gives $n = 17$.
3	**(b)** 19	AP with $a = 4$, $d = 3$: $t_6 = 4 + 5 \times 3$.
4	**(d)** 47	$a + (n-1)d = 5 + 14 \times (3)$.
5	**(a)** 210	$S_n = \frac{n}{2}[2a + (n-1)d]$ with $a = 3, d = 4$.
6	**(d)** 162	$ar^{n-1} = 2 \times 3^4$.
7	**(a)** 21	$3 + (n-1)4 = 83$ gives $n = 21$.
8	**(d)** an arithmetic progression	Definition of an AP.
9	**(b)** 44	$3 \times 16 - 4$.
10	**(c)** 21th term	$4n - 3 = 81$ gives $n = 21$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$t_8 = 25$ (1); $3n + 1 = 37$ gives $n = 12$, so yes (1).	1+1
12	Differences $= 5$, constant (1); $t_{12} = 8 + 11 \times (5) = 63$ (1).	1+1
13	$t_2 = 3$ (1); Terms $1, 3, 7, 15, 31$ (1).	1+1
14	$t_7 = 33$ (1); $5n - 2 = 43$ gives $n = 9$, so yes (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$7 + (n-1)(6) = 151$ (1); $(n-1) \times (6) = 144$ (1); $n = 25$ (1).	1+1+1
16	$a = 6$, $d = 5$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{15} = 615$ (1).	1+1+1
17	$r = 4$ (1); $t_n = ar^{n-1} = 3 \times 4^4$ (1); $t_5 = 768$ (1).	1+1+1
	OR: $90 + (n-1)(-5) = 25$ (1); $(n-1) \times (-5) = -65$ (1); $n = 14$ (1).	1+1+1
18	$a = 25$, $d = -3$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{10} = 115$ (1).	1+1+1
	OR: $r = 5$ (1); $t_n = ar^{n-1} = 2 \times 5^3$ (1); $t_4 = 250$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Fares ₹240, ₹280, ₹320 (1); Common difference ₹40, so an AP (1); $t_n = a + (n-1)d$ with $a = 240$, $d = 40$ (1); $t_{12} = 240 + 11 \times 40$ (1); Fare ₹680 (1).	1+1+1+1+1
	OR: Monthly savings form an AP (1); $t_{12} = 300 + 11 \times 50 = 850$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{12} = \frac{12}{2}[600 + 550]$ (1); Total ₹6900 (1).	1+1+1+1+1
20	Fares ₹175, ₹200, ₹225 (1); Common difference ₹25, so an AP (1); $t_n = a + (n-1)d$ with $a = 175$, $d = 25$ (1); $t_{18} = 175 + 17 \times 25$ (1); Fare ₹600 (1).	1+1+1+1+1
	OR: Monthly savings form an AP (1); $t_{10} = 500 + 9 \times 20 = 680$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{10} = \frac{10}{2}[1000 + 180]$ (1); Total ₹5900 (1).	1+1+1+1+1

Mistakes to avoid:

- Writing $t_n = a + nd$ instead of $a + (n-1)d$, which makes the first term wrong.
- Forgetting the factor $\frac{n}{2}$ in S_n, or using $n(n+1)$ for the sum of the first n natural numbers (it is $\frac{n(n+1)}{2}$).
- Using $t_n = ar^n$ for a GP instead of ar^{n-1}.
- Calling a sequence an AP after checking only the first two terms; the difference must be the same for all consecutive terms.
- Taking S_n of an AP to be the nth term, or finding the nth term as S_n instead of $S_n - S_{n-1}$.
- Sign errors with a negative common difference or ratio, e.g. $t_5 = 7 + 4 \times (-2) = -1$, not $7 + 4 \times 2$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Sequences and Progressions** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 29	AP with $a = 9$, $d = 4$: $t_6 = 9 + 5 \times 4$.
2	**(b)** -4	$a + (n-1)d = 40 + 11 \times (-4)$.
3	**(d)** 282	$S_n = \frac{n}{2}[2a + (n-1)d]$ with $a = 7, d = 3$.
4	**(b)** 192	$ar^{n-1} = 3 \times 2^6$.
5	**(d)** 26	$5 + (n-1)7 = 180$ gives $n = 26$.
6	**(c)** a geometric progression	Definition of a GP.
7	**(c)** 88	$1 \times 81 + 7$.
8	**(b)** 14th term	$6n + 5 = 89$ gives $n = 14$.
9	**(c)** 5	AP with $a = 15$, $d = -2$: $t_6 = 15 + 5 \times -2$.
10	**(b)** 89	$a + (n-1)d = -6 + 19 \times (5)$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Differences $= -4$, constant (1); $t_9 = 30 + 8 \times (-4) = -2$ (1).	1+1
12	$t_2 = 7$ (1); Terms $2, 7, 17, 37, 77$ (1).	1+1
13	$t_6 = 29$ (1); $4n + 5 = 65$ gives $n = 15$, so yes (1).	1+1
14	Differences $= 8$, constant (1); $t_{15} = 1 + 14 \times (8) = 113$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$4 + (n-1)(7) = 214$ (1); $(n-1) \times (7) = 210$ (1); $n = 31$ (1).	1+1+1
16	$a = 9$, $d = 8$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{18} = 1386$ (1).	1+1+1
17	$r = 3$ (1); $t_n = ar^{n-1} = 6 \times 3^5$ (1); $t_6 = 1458$ (1).	1+1+1
	OR: $150 + (n-1)(-8) = 6$ (1); $(n-1) \times (-8) = -144$ (1); $n = 19$ (1).	1+1+1
18	$a = 40$, $d = -6$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{11} = 110$ (1).	1+1+1
	OR: $r = 3$ (1); $t_n = ar^{n-1} = 1 \times 3^7$ (1); $t_8 = 2187$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Fares ₹115, ₹130, ₹145 (1); Common difference ₹15, so an AP (1); $t_n = a + (n-1)d$ with $a = 115$, $d = 15$ (1); $t_{30} = 115 + 29 \times 15$ (1); Fare ₹550 (1).	1+1+1+1+1
	OR: Monthly savings form an AP (1); $t_{16} = 100 + 15 \times 30 = 550$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{16} = \frac{16}{2}[200 + 450]$ (1); Total ₹5200 (1).	1+1+1+1+1
20	Fares ₹72, ₹84, ₹96 (1); Common difference ₹12, so an AP (1); $t_n = a + (n-1)d$ with $a = 72$, $d = 12$ (1); $t_{25} = 72 + 24 \times 12$ (1); Fare ₹360 (1).	1+1+1+1+1
	OR: Monthly savings form an AP (1); $t_{20} = 250 + 19 \times 10 = 440$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{20} = \frac{20}{2}[500 + 190]$ (1); Total ₹6900 (1).	1+1+1+1+1

Mistakes to avoid:

- Writing $t_n = a + nd$ instead of $a + (n-1)d$, which makes the first term wrong.
- Forgetting the factor $\frac{n}{2}$ in S_n, or using $n(n+1)$ for the sum of the first n natural numbers (it is $\frac{n(n+1)}{2}$).
- Using $t_n = ar^n$ for a GP instead of ar^{n-1}.
- Calling a sequence an AP after checking only the first two terms; the difference must be the same for all consecutive terms.
- Taking S_n of an AP to be the nth term, or finding the nth term as S_n instead of $S_n - S_{n-1}$.
- Sign errors with a negative common difference or ratio, e.g. $t_5 = 7 + 4 \times (-2) = -1$, not $7 + 4 \times 2$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Sequences and Progressions** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 260	$S_n = \frac{n}{2}[2a + (n-1)d]$ with $a = 50, d = -5$.
2	**(a)** 125	$ar^{n-1} = 1 \times 5^3$.
3	**(d)** 18	$100 + (n-1) - 6 = -2$ gives $n = 18$.
4	**(a)** the sum of the two terms before it	Recursive rule.
5	**(c)** 43	$5 \times 9 - 2$.
6	**(b)** 30th term	$7n - 1 = 209$ gives $n = 30$.
7	**(b)** 36	AP with $a = 1$, $d = 7$: $t_6 = 1 + 5 \times 7$.
8	**(d)** 83	$a + (n-1)d = 11 + 8 \times (9)$.
9	**(b)** 632	$S_n = \frac{n}{2}[2a + (n-1)d]$ with $a = 2, d = 5$.
10	**(b)** 160	$ar^{n-1} = 5 \times 2^5$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$t_2 = 5$ (1); Terms $3, 5, 9, 17, 33$ (1).	1+1
12	$t_{10} = 29$ (1); $2n + 9 = 31$ gives $n = 11$, so yes (1).	1+1
13	Differences $= 6$, constant (1); $t_{11} = -3 + 10 \times (6) = 57$ (1).	1+1
14	$t_2 = 7$ (1); Terms $1, 7, 19, 43, 91$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$11 + (n-1)(12) = 203$ (1); $(n-1) \times (12) = 192$ (1); $n = 17$ (1).	1+1+1
16	$a = 1$, $d = 3$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{25} = 925$ (1).	1+1+1
17	$r = 2$ (1); $t_n = ar^{n-1} = 4 \times 2^8$ (1); $t_9 = 1024$ (1).	1+1+1
	OR: $3 + (n-1)(10) = 393$ (1); $(n-1) \times (10) = 390$ (1); $n = 40$ (1).	1+1+1
18	$a = 13$, $d = 4$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{14} = 546$ (1).	1+1+1
	OR: $r = 3$ (1); $t_n = ar^{n-1} = 5 \times 3^4$ (1); $t_5 = 405$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Fares ₹98, ₹116, ₹134 (1); Common difference ₹18, so an AP (1); $t_n = a + (n-1)d$ with $a = 98$, $d = 18$ (1); $t_{22} = 98 + 21 \times 18$ (1); Fare ₹476 (1).	1+1+1+1+1
	OR: Monthly savings form an AP (1); $t_{14} = 400 + 13 \times 25 = 725$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{14} = \frac{14}{2}[800 + 325]$ (1); Total ₹7875 (1).	1+1+1+1+1
20	Fares ₹150, ₹180, ₹210 (1); Common difference ₹30, so an AP (1); $t_n = a + (n-1)d$ with $a = 150$, $d = 30$ (1); $t_{15} = 150 + 14 \times 30$ (1); Fare ₹570 (1).	1+1+1+1+1
	OR: Monthly savings form an AP (1); $t_{24} = 50 + 23 \times 5 = 165$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{24} = \frac{24}{2}[100 + 115]$ (1); Total ₹2580 (1).	1+1+1+1+1

Mistakes to avoid:

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